Excessive Heat and Correctional Officer Turnover: Evidence from Texas

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Abstract

Correctional work is difficult, dangerous, and hot. Many U.S. prisons lack air conditioning, resulting in sweltering conditions for both incarcerated people and the street-level bureaucrats tasked with their supervision. We examine the effect of working in the summer heat without air conditioning on correctional officer turnover using personnel records of 62,164 Texas Department of Criminal Justice (TDCJ) employees across 70 prisons between 2010 and 2023. We demonstrate that during the hottest months of the year, correctional employees in prisons lacking air conditioning are approximately 40% more likely to leave their positions than employees in air-conditioned prisons experiencing similar summer temperatures. The moderating effect of air conditioning dissipates as temperatures cool and is isolated to newer frontline correctional officers who are likely less acclimated to the heat. Finally, back-of-the-envelope estimates suggest that installing air conditioning to combat understaffing is comparable in cost to raising salaries, the TDCJ's standard response to turnover.

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Prisons and jails across the U.S. are contending with an understaffing crisis. Hiring and retaining correctional officers has long been difficult (Lambert 2001), but rising incarceration rates and the COVID-19 pandemic have brought many prison systems to the tipping point. Between 2019 and 2022, the number of state-employed correctional officers declined by 10%, with states like Arkansas and Georgia experiencing declines of 20% and 29%, respectively (Heffernan and Li 2024). In some states, vacancy rates among correctional staff are as high as 50% despite widespread initiatives to attract new hires (Lieb 2023). Reduced staffing makes prisons more dangerous for both employees and incarcerated people. Numerous accounts document how understaffing is associated with increased violence, staff assaults, prolonged lockdowns, and poor health outcomes for incarcerated people (Heffernan and Li 2024; Lambert 2001; Minor et al. 2011; Nam-Sonenstein and Sanders 2024).

While particularly extreme in U.S. prisons, understaffing and turnover are common issues in positions occupied by street-level bureaucrats. Across sectors, street-level bureaucrats face common challenges, including high caseloads, insufficient time with clients, and feelings of alienation from their work (Lipsky 2010). As such, stress and job burnout are frequently reported among both correctional officers and other types of street-level bureaucrats as key reasons for departing the workforce (Cullen et al. 1985; Finney et al. 2013; Lambert 2001; Schwartz et al. 2024; Shim, Park, and Eom 2017). In addition to these challenges, street-level bureaucrats also often face difficult physical working environments. Many state and federal government buildings across the country have considerable maintenance backlogs (Marroni 2025; *Beyond the Backlog* 2025), and agencies often lack the political or financial capital to improve working conditions (Wilson 2019). Government employees desire jobs with, among other criteria, clearly-defined roles where they are treated with respect and included in organizational decision making (Hur and Abner 2024), but they likely also prioritize working somewhere safe and comfortable.

We assess the impact of poor working conditions on turnover among street-level bureaucrats by focusing on a specific, increasingly relevant case: the effect of excessive heat on correctional officer turnover. Across the U.S., prisons are confronting unprecedented levels of heat (Tuholske et al. 2024; Wu and Felder 2022). Very few state prison systems have

universal air conditioning, resulting in sweltering conditions in the summer months, with some prisons reaching indoor temperatures of over 130°F (Salhotra and Melhado 2024). The poor conditions brought about by excessive heat have spurred numerous lawsuits from incarcerated people, protests from correctional officer organizations, and instigated a growing body of research documenting the negative effects of heat on prison violence (Mukherjee and Sanders 2021) and incarcerated individuals' health (Skarha et al. 2022; Skarha et al. 2023).

Texas's state prison system is an ideal setting for evaluating heat's role in fostering poor working conditions for correctional officers. The Texas Department of Criminal Justice (TDCJ) operates the largest state prison system in the country. In August 2023, the agency employed over 17,000 correctional officers and supervised 130,000 incarcerated individuals across nearly 100 facilities (*Sunset Report* 2025). Despite Texas prisons being exposed to high levels of extreme heat in the summer months (Tuholske et al. 2024), only 30 of the 105 TDCJ facilities that have been in use at some point between 2010 and 2023 have fully air-conditioned housing areas. Twenty-three facilities lack any air conditioning in their dormitories, with the remaining prisons having partial, often minimal, coverage. As a result, conditions within many TDCJ prisons are brutal in the summer months. A federal judge in 2025 called the lack of air conditioning in TDCJ prisons "plainly unconstitutional," while a different federal judge wrote in a 2018 injunction that incarcerated people in Texas "face a substantial risk of serious harm from the sweltering Texas heat" (McCullough 2017; Salhotra 2025). Union officials in the state have called excessively hot Texas prisons "death traps" and likened working in units without air conditioning to "going up and down stairs at a football stadium in the heat of the day, while wearing a coat" (Dang 2022; Martin 2013).

We test whether these conditions lead correctional officers to depart the TDCJ workforce using individual-level personnel data and facility-level air-conditioning data. The personnel records consist of monthly snapshots of all TDCJ staff employed in security positions across the state from January 2010 to January 2023. The records are rich, noting employees' full names, race, sex, ages, job titles, salaries, hire dates, and employing facilities. In order to determine whether correctional officers work in air-conditioned facilities, we rely on data from the Texas Prisons Community Advocates (TPCA), a nonprofit advocacy organization. The

facility-level air-conditioning data, which the TPCA received via public-records requests to the TDCJ, note whether each prison's housing areas have full, partial, or no air conditioning.

Building upon previous research that has estimated the effects of heat within carceral settings (e.g., Mukherjee and Sanders 2021), we utilize seasonal variation in temperature to estimate the effect of excessive heat on correctional officer turnover. In particular, we utilize seasonal temperature swings, combined with variation in the presence of air conditioning, to examine the effects of working in hot, uncooled conditions. Our primary analysis tests whether employees are more likely to depart in the hottest months of the year, relative to the coldest, if they work in a prison lacking air conditioning. We demonstrate that, before the onset of the COVID-19 pandemic, correctional officers in prisons without air conditioning were approximately 40% more likely to depart in the hottest months of the year, relative to their peers in air-conditioned facilities experiencing similar temperatures. We demonstrate that the moderating effect of air conditioning is isolated to newer frontline correctional officers who are likely less acclimated to working in hot temperatures. There is no difference in the probability of departing in the summer, relative to the winter, among frontline employees with six or more years of experience across prisons with and without air conditioning. We also show via a placebo test that differences in turnover across prisons with and without air conditioning dissipate as temperatures cool, underscoring that the extreme temperatures resulting from a lack of air conditioning are likely driving employees' increased propensity to depart.

Our results show how prison cooling can alleviate understaffing, but objections to installing air conditioning frequently hinge on financial concerns. To evaluate whether air conditioning is a cost-effective policy, we conduct back-of-envelope cost analyses of both installing air conditioning in TDCJ facilities and the agency's de facto policy for addressing employee turnover: raising salaries. On average across various cost models, we estimate that installing air conditioning throughout the TDCJ system costs $56 million annually. Raising employee salaries by 7.6%, the amount necessary to reduce turnover by the same amount as installing air conditioning, costs $81 million per year. Under all but the most pessimistic scenarios, adding air conditioning to TDCJ facilities is similar in cost or more cost-effective than

a commensurate increase in employee salaries. More broadly, this suggests that investments in workplace infrastructure may yield greater returns on investment than compensation-based retention strategies alone. This result fits with research on turnover among other street-level bureaucrats that emphasizes the positive but limited effects of financial compensation on improving retention (e.g., Nguyen et al. 2019).

Overall, our results demonstrate that the poor working conditions brought about by excessive heat contribute to heightened turnover among correctional officers, a sizable and consequential group of street-level bureaucrats. We also show that air-conditioning offers an effective, albeit partial remedy. In addition to improving the health and well-being of incarcerated people, air conditioning enhances working conditions for correctional staff, increasing retention and reducing the dangerous conditions that emerge when prisons lack adequate staffing. However, air conditioning addresses just one of many issues within correctional institutions. While effectual, installing air conditioning does not fully address other issues, such as violence and overcrowding, that likely also contribute to understaffing in American prisons.

1 Literature Review

Often tasked with considerable responsibilities with limited resources, teachers, police officers, correctional officers, social services workers, and other types of frontline public sector employees face common stresses and challenges (Lipsky 2010). Across sectors, these street-level bureaucrats contend with high levels of stress and burnout, largely due to the nature of the work performed and the degree of organizational support (Drew, Sargeant, and Martin 2024; Finney et al. 2013; Ingersoll 2001; McCarty and Skogan 2013; Radey and Wilke 2023). As such, research has documented high levels of turnover among these employees (e.g., Nguyen et al. 2019; Russo 2019; *The State of Recruitment & Retention* 2025; Lushin et al. 2023). Likewise, despite the differences across street-level roles, research on improving employee retention points to similar solutions, including increasing financial compensation, improving experiences with supervisors, increasing administrative support, and making workloads more manageable (Nguyen et al. 2019; Farahmandpour and Voelkel 2025;

Boyd et al. 2011; Radey and Wilke 2023).

It should not come as a surprise that departure rates are particularly high among correctional officers. Colloquially described as "the toughest beat" (Page 2011), correctional work is dangerous, low-paying, and stressful. A survey of state correctional officers in California found that 70% of respondents had witnessed someone be seriously injured or killed, while 18% reported being seriously injured at work (Lerman, Harney, and Sadin 2022). Correctional workers are also increasingly being asked to compensate for staff shortages by working longer hours with less support. In Georgia, the state increased overtime spending for correctional workers by over 1,000% between 2019 and 2022, while the size of the correctional workforce decreased by approximately 33% (Heffernan and Li 2024). The snowballing effects of understaffing recently came to a head in New York when the commissioner of the state's prisons system wrote in a memo that "70% of our original staffing model is the new 100%," which helped to instigate large-scale unauthorized strikes among dissatisfied correctional employees (Roebuck 2025).

The numerous challenges of correctional work take a toll on correctional officers' physical and mental health. Correctional officers are at heightened risk of anxiety, Post-Traumatic Stress Disorder, burnout, job dissatisfaction, and COVID-19 infection (Cheeseman et al. 2011; Cullen et al. 1985; Finney et al. 2013; Lambert 2001; Schwartz et al. 2024; Ward et al. 2021). Several studies have also indicated that the severe stress and burnout tied to correctional work are associated with higher departure rates among officers (Griffin, Hogan, and Lambert 2014; Lerman, Harney, and Sadin 2022; Vickovic, Morroe, and Lambert 2022). Perhaps most disconcerting is the elevated risk of suicide among correctional officers. One study estimated that the suicide rate is 39% higher among correctional officers than the general public (Stack and Tsoudis 1997), and more recent research suggests that elevated suicide rates persist (Frost and Monteiro 2020; Lerman, Harney, and Sadin 2022).

The difficulties of correctional work have prompted investigations into how support from administrators and peers can decrease turnover. Studies have shown that facilitating higher organizational commitment among correctional officers, improved support from prison lead-

ers, and higher-quality supervisors all improve retention (Lambert and Hogan 2009; Lambert 2001; Mitchell et al. 2000; Butler et al. 2019; Griffin, Hogan, and Lambert 2014; Leip and Stinchcomb 2013). Likewise, employees who express satisfaction with their coworkers are also at less risk of leaving (Minor et al. 2011). Yet, it is not only correctional officers' relationships with their colleagues and supervisors that influence their experiences on the job. The physical conditions of their workplace likely matter too. Research on teachers, for instance, has shown that poor building quality is associated with increased departure rates (Buckley, Schneider, and Shang 2005; Loeb, Darling-Hammond, and Luczak 2005). Numerous reports have pointed to the dilapidated condition of the state prisons where many incarcerated people live and correctional officers work. A recent report contracted by the Florida Department of Corrections, for example. suggested that 37% of the state's prisons were in poor or critical condition (KPMG 2023). A similar report in Illinois highlighted $2.5 billion in deferred maintenance needs, while another report suggested that California invest $11 billion in 12 of the state's oldest prisons (O'Neil 2020; *Illinois Facility Master Plan* 2023).

The physical aspects of correctional officers' day-to-day experiences on the job likely influence their willingness to remain in their roles. However, existing work on correctional officer turnover makes it difficult to parse the role of working conditions on turnover. Much of the existing literature is exploratory or relies on surveys fielded to a small number of correctional officers (Lambert and Hogan 2009; Lambert 2001; Mitchell et al. 2000; Butler et al. 2019; Griffin, Hogan, and Lambert 2014; Leip and Stinchcomb 2013). It is difficult to generalize findings from a survey of a few hundred correctional officers in a single prison or correctional system, especially when the variable of interest—working conditions—is unlikely to vary at the employee level. Understanding the impact of employees' physical working conditions, which likely underlie the psychological states scholars have shown to correlate with an increased propensity to exit, requires observing many correctional officers experiencing different working conditions over time.

2 Heat in Prisons

Exposure to extreme heat is a severe and increasingly salient case of poor working conditions in U.S prisons and beyond. Considerable research has identified negative social outcomes of heat exposure. Gun violence and other forms of violent behavior are more prevalent on high-temperature days (Burke, Hsiang, and Miguel 2015; Colmer and Doleac 2023; Hsiang, Burke, and Miguel 2013). A meta-analysis of studies testing the effect of climate on conflict finds that a one standard deviation increase in temperature increases the frequency of interpersonal violence and intergroup conflict by 4% and 14%, respectively (Hsiang, Burke, and Miguel 2013). Goodman et al. (2018) show that heat exposure also leads to educational deficits, as warmer school days diminish learning and testing outcomes. Other studies have shown that heat negatively impacts economic productivity, civic engagement, and health (Deschenes 2014; Lamare 2013; Graff Zivin, Hsiang, and Neidell 2018).

Extreme heat is a persistent and largely unaddressed issue in many U.S. correctional facilities. Forty-four states lack universal air conditioning, including most Southern states where dangerous prison temperatures are most common (Tuholske et al. 2024). Temperatures in these non-air-conditioned prisons can exceed 130°F and have been likened to "torture chambers," "concrete coffins," and "sitting inside of a convection oven" (Lartey 2023; Salhotra and Melhado 2024). Incarcerated individuals sometimes adopt extreme measures to survive in these conditions, including flooding their cells to lie in the wet concrete, lighting fires, bathing in toilet water, or screaming in unison for help (Goodman 2023; *Wilson v. Dixon,* n.d.).

Excessive heat in prisons has been linked to numerous negative consequences for incarcerated individuals, including higher rates of mortality, violence, suicide attempts, and health problems (Cloud et al. 2023; Lartey 2023; Mukherjee and Sanders 2021; Skarha et al. 2023; Tuholske et al. 2024). For example, Mukherjee and Sanders (2021) leverage daily variation in temperature to study the effect of heat on violence in Mississippi prisons. They find that days with unsafe heat indices increase the number of daily violent interactions by 20%. Using data from six state prisons in Louisiana, Cloud et al. (2023) also show a strong positive asso-

ciation between high-temperature days and suicide-watch incidents. Another study suggests that roughly 13% of all deaths during warm months in Texas prisons may be attributable to extreme heat (Skarha et al. 2022). While southern states have been the focus of many studies of the negative effects of heat within prisons, rising temperatures due to climate change will likely make heat more of a salient issue in other parts of the country too. Individuals living and working in prisons in the Pacific Northwest, Midwest, and New England are less accustomed to hazardous heat levels, potentially increasing the dangers of rising temperatures in facilities without effective heat mitigation.

Excessive heat is a highly relevant concern for correctional staff. Between 2012 and 2013, 147 state correctional employees in Texas reported illness or injury due to heat (Martin 2013). Incarcerated people in the state have also reported seeing correctional workers faint on account of heat (*Deadly Heat in U.S. (Texas) Prisons* 2024). Despite the lack of scholarly work on the effects of excessive heat on correctional staff, some anecdotal evidence suggests that it also impacts officers' ability to do their jobs. For instance, in New Jersey, correctional workers have reported that excessive heat negatively affected their incident response times (Schuster and King 2022). Even in the absence of major health and safety concerns, excessive heat makes correctional work less appealing. For example, the head of the Louisiana Department of Public Safety and Corrections noted that officers in prisons without air conditioning were changing their clothes up to three times per shift (O'Donoghue 2022).

A number of anecdotal reports suggest that the challenges faced by employees working in excessively hot conditions negatively impact correctional officer retention (Ballard 2022; McGivern 2024; Salhotra 2024). There are several pathways through which extreme heat may contribute to correctional officer turnover. Most directly, correctional workers might be more likely to depart in search of jobs with better working conditions during the summer months when heat is particularly salient. However, employees may leave in the spring in anticipation of the summer heat. Of course, they may also depart in the fall after working in extreme during the prior summer. Heat may also operate more indirectly by influencing the selection of correctional officers. Stronger applicants who are less likely to depart may opt out of working in correctional facilities without air conditioning, leaving prisons to be staffed

by employees with higher baseline probabilities of departing. Given that correctional officers already exhibit volatile employment histories and short tenure spells (Burton et al. 2022), any or all of these mechanisms may plausibly trigger departures among a workforce frequently on the brink of quitting.

While it is difficult to fully parse these different channels empirically, they do provide some testable expectations. Namely, employees in prisons without air conditioning are more likely to depart than their peers in cooled facilities. This will especially, although not exclusively, be the case in the hottest months of the year when heat is most salient to employees. Furthermore, newer frontline correctional officers with less experience are also likely more affected by working in excessively hot conditions. Unlike their peers who have been on the job for a few summers, less-experienced correctional officers have not become acclimated to the hot conditions. Research on heat acclimatization shows that sustained exposure to high heat alters both physiological and subjective markers of heat tolerance (Malgoyre et al. 2020). Acclimatization can occur quickly when exposure is consistently intense (Khalaf, Ramadan, and Al-Ashaikh 2017; Saillant, Charkoudian, and Salgado 2022). Correctional officers in prisons without air conditioning, however, are exposed to extreme temperatures periodically depending on their posts, daily duties, and shift times, which likely extends the time required to acquire heat tolerance. As a result, employees who have experienced multiple summers of high temperatures in prisons without air conditioning will likely become increasingly tolerant of the heat, leaving newer frontline employees as the most likely to leave during hot times of the year.

3 Data

We collected monthly snapshots of all correctional staff employed by the TDCJ from January 2010 through January 2023 via multiple public records requests to the agency. For each month, the data note the full name, age, facility, race, sex, full-time status, job title, and most recent hire date for all correctional staff employed in security positions at TDCJ facilities. Most employees in the dataset are frontline correctional officers (83%), but the data also include first-line supervisors, such as sergeants, lieutenants, and captains, as well

as managers (e.g., majors, assistant wardens, and wardens). The data also include some employees in auxiliary roles, like food and laundry managers. The personnel files do not include unique employee identification codes that follow employees throughout their tenure with the TDCJ. As a result, we assign unique employee identification codes by grouping together observations with similar ages and the same first name, middle initial, last name, race, last hire date, and sex. Since the codes include employees' last hire date with the TDCJ, they can capture instances where the same employee departs the workforce and then returns later in the study period.[1]

We determine whether correctional employees work in air-conditioned facilities using data published by the TPCA, a nonprofit advocacy organization. Through public records requests, the TPCA gathered data on whether TDCJ prisons' housing areas were fully, partially, or not air-conditioned as of April 2022. These air conditioning data only pertain to the housing areas where incarcerated people live and frontline correctional officers frequently work. According to the TDCJ, all prisons' administrative, educational, and medical areas have air conditioning, even if housing areas are uncooled. For prisons with partial air conditioning, the data note what share of beds in the prisons are cooled via air conditioning. Figure 1 shows the distribution of air-conditioning coverage in TDCJ prisons as of April 2022. Most prisons in the state are either fully air-conditioned or have air conditioning covering 10% or fewer of their beds.

A key concern with the TPCA data is that it is static. An ideal test of the effects of extreme heat on correctional officer turnover would likely utilize over-time changes in the number of air-conditioned beds in a prison. However, such an ideal test is impractical. For one, the realities of the construction process limit the efficacy of over-time installation data. Many states, like Texas, are installing air conditioning units at a slow rate—if at all—limiting the strength of a potential longitudinal treatment measure. Moreover, installing air conditioning takes a significant amount of time, reducing any potential unanticipated shock from the introduction of additional cooled beds. Data availability further complicates using longitudinal measures of air conditioning coverage. Texas' open record laws allow agencies to

1. Thirteen percent of employees in our dataset had more than one employment spell with the TDCJ.

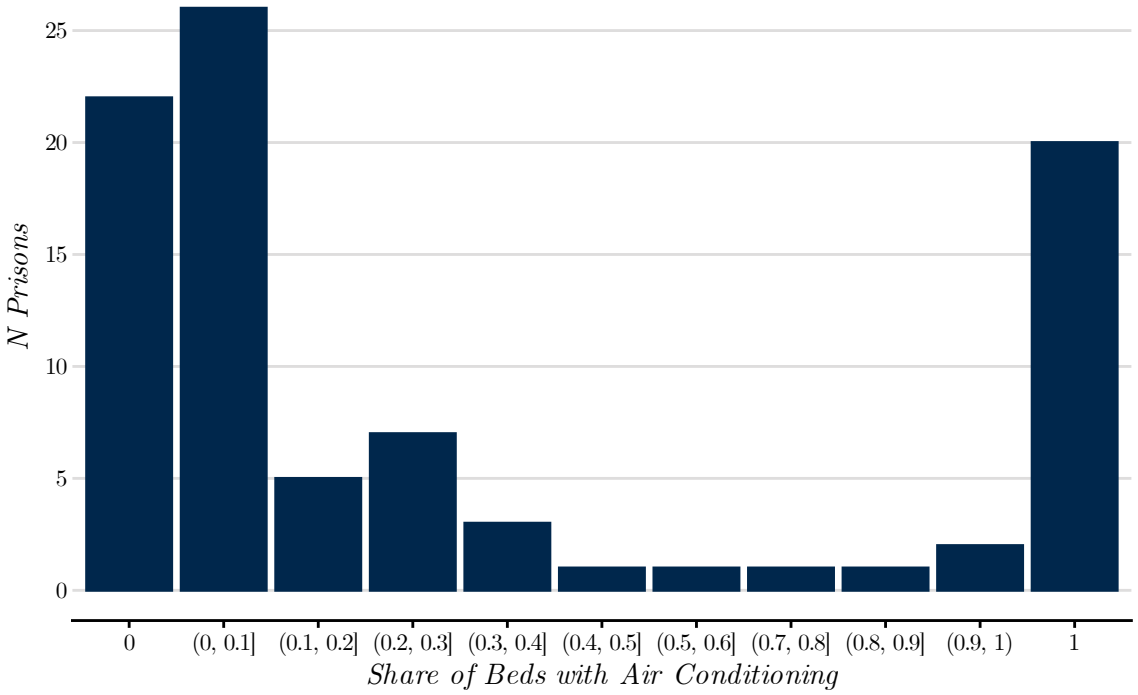
exclude data related to ongoing litigation from release. Given the frequency of heat-related lawsuits brought by individuals incarcerated by the TDCJ, this is an effective check, for the time being, on collecting the additional older data necessary to cover our entire study period.[2]

Despite the challenges with using static measures of air conditioning coverage — which we also address below in the Methods section — we believe these data from the TPCA are a solid measure of prisons' air conditioning coverage over most of our study period. For one, the TDCJ has been slow to install air-conditioned beds in their facilities. As of January 1, 2026, the TDCJ has installed over 14,000 cooled beds since 2018, but most of those installations occurred during or after 2022. Between 2018 and 2021, the agency constructed less than 1,000 air-conditioned beds per year (Betts 2025). As a result, only a few thousand installations occurred between 2018 and the date of the TPCA air conditioning snapshot. While it is not possible to get an accurate count of installations of air conditioning before 2018, we have not found any evidence to suggest the TDCJ was proactive about cooling prisons over the earlier years of our study period. As a result, our air conditioning measure likely slightly overestimates air conditioning coverage in earlier years, which works against our expectation of increased turnover in prisons lacking air conditioning.

Given the distribution of air conditioning coverage in Figure 1, we opt to binarize the continuous measure of air conditioning in TDCJ prisons. Since most prisons have either all or 10% or fewer of their beds cooled, we opt to define a prison as having air conditioning if 90% or more of its beds are air conditioned. Likewise, a prison is deemed to not have air conditioning if 10% or fewer of its beds are cooled. While this approach balances including as much data as possible while retaining a meaningful difference across the two groups of prisons, we also show that our main results are largely robust with two alternative definitions. The first uses employee data from all prisons by defining a prison as (not) having air conditioning if more (less) than 50% of beds are cooled. The second only uses prisons with housing areas that have full or no air conditioning. A key advantage of this latter approach is that prisons with

2. Recent court cases have pushed the TDCJ to publish snapshots every few months of their progress in installing air conditioning in prisons. However, these publications only cover recent installations and do not include data over most of our study period.

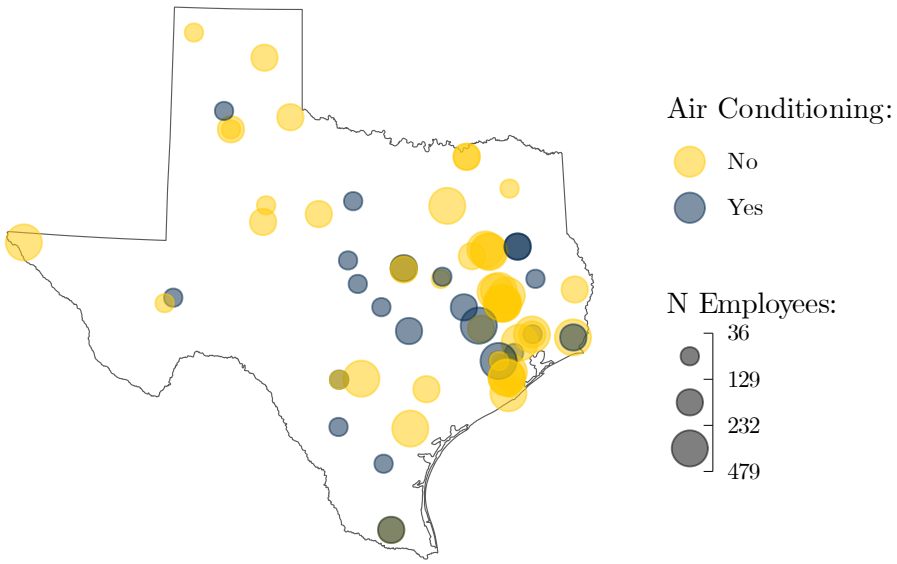
Figure 1 – Distribution of Air-Conditioning Coverage in Texas Prisons (2022). Shows the distribution of the share of beds in Texas prisons that have air conditioning. Data is from the TPCA.



full or no air conditioning are even less susceptible to over-time changes in air-conditioning coverage. We are only aware of two prisons, the Hodge and Pack Units, that had full air conditioning installed during our study period after previously lacking any air conditioning in housing areas. These installations followed a four-year legal battle between individuals incarcerated at the Pack Unit and the state, underscoring that the TDCJ rarely makes full-scale changes to prisons' air conditioning (McCullough 2017, 2018a; Blakinger and Banks 2018).[3]

3. An archived version of a TDCJ dashboard tracking how many "cool beds" were installed between 2018 and October 2023 suggests that 5,876 beds were installed over the period. Besides the 2,467 cool beds installed in the Hodge and Pack units, only 26 air-conditioned beds were installed in a facility (Hilltop Unit) noted as having no air conditioning in the TPCA data from April 2022 ("TDCJ Air Conditioning Construction Projects" 2023). A more recent snapshot of facility air-conditioning status provided by the TPCA dated August 2024 suggests that only fourteen TDCJ facilities now have no air-conditioning coverage, usually due to the installation of 50 or fewer cool beds.

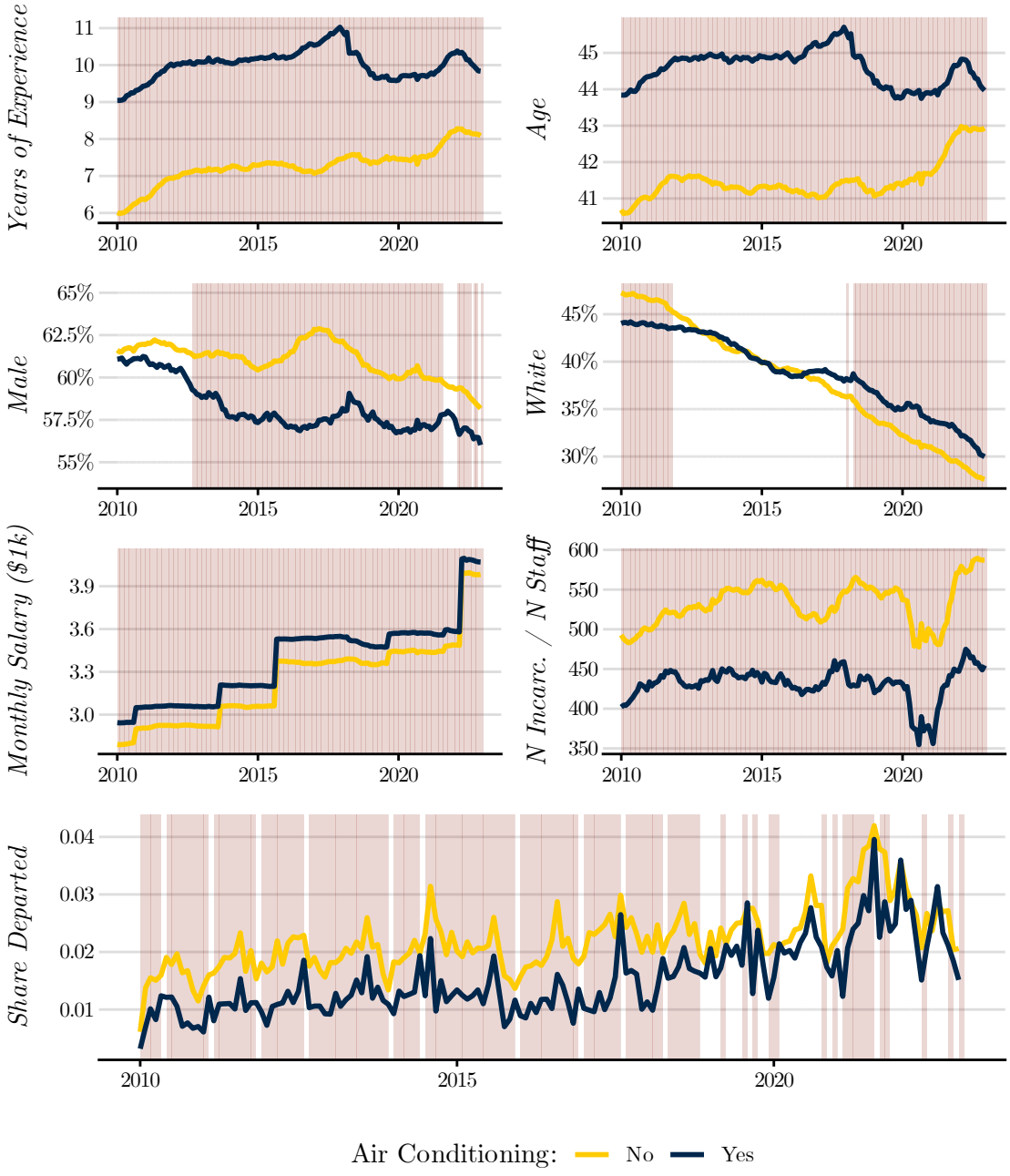
Figure 2 – Location of TDCJ Prisons by Air-Conditioning Status and Number of Employees. Shows the location of the seventy prisons operated by the TDCJ at some point between 2010 and 2023 with $> 90\%$ or $< 10\%$ air conditioning coverage in housing areas. Points are binned into tertiles based on the number of employees working in the facility in the last month for which data is available. For most prisons, the last observed month is December 2022. Two units, Jester I and Scott, were closed in 2020 and employee counts are therefore pulled from August and December 2020, respectively. The Hodge and Pack units are shown as having full air conditioning.



The resulting dataset covering the seventy prisons with air conditioning coverage of $> 90\%$ or $< 10\%$ includes 62,164 distinct employees, 69,994 employment spells, and 2,563,528 employee-month observations. Figure 2 shows the location of these prisons across the state. Air-conditioned facilities employ fewer correctional staff and are slightly more concentrated in the western part of the state. The employees in these facilities also differ on other dimensions. Figure 3 shows descriptive characteristics of TDCJ correctional staff over the course of our study period. Monthly group means for employees in facilities with and without air conditioning are shown in blue and yellow, respectively, while red highlighting indicates that the null hypothesis of no difference in means is rejected using a two-sided t-test ($p<.05$). Employees in facilities without air conditioning are significantly younger with fewer years of experience and lower earnings than their peers in air-conditioned facilities. They are also more likely to be male and, in more recent years, less likely to be white. Incarcerated-person-to-staff ratios are also higher in facilities without air conditioning, underscoring both the difficulty

of staffing these positions and the difficult conditions faced by remaining employees.

Figure 3 – Correctional Staff Differ Across Prisons With and Without AC. Shows the mean years of experience, age, monthly salary, share male, share white, incarcerated person/staff ratio, and departure rate in TDCJ facilities with and without air conditioning by month. Data includes both frontline and supervisory staff. Time periods are highlighted in red if the null hypothesis of no difference in means is rejected using a two-sided t-test ($p<.05$).



Our main outcome of interest is whether an employee departs the TDCJ workforce in a given month.[4] It encompasses both instances where an employee leaves the TDCJ workforce and does not return during our study period as well as cases of an employee leaving before later returning to correctional work.[5] The bottom facet of Figure 3 shows the share of TDCJ correctional employees that depart each month from prisons with and without air conditioning. The panel highlights a few key trends. First, up to the onset of the COVID-19 pandemic, departures were more common in prisons without air conditioning. Second, departures from prisons with air conditioning began to increase around 2017. As a result, by the time the COVID-19 pandemic began, the difference in departures rates from prisons with and without air conditioning had considerably narrowed. Finally, employee departures from both types of prisons spike following the onset of the pandemic in March 2020. Departure rates peak in summer and fall of 2021 and by the beginning of 2022 are largely similar across both types of prisons.

4 Methods

Following the approach taken by Mukherjee and Sanders (2021), we take advantage of seasonal variation in temperature to identify the effect of excessive heat on correctional officer turnover. Heat likely has the largest impact on correctional officers in the summer months. Departures are still possible in the spring or fall due to the heat as employees anticipate or react to working in the summer, but heat is most salient when it is hot out. Likewise, employees are likely least sensitive to hot temperatures during the winter when air conditioning is largely irrelevant. Therefore, the coldest months of the year serve as a useful control for measuring employees' baseline propensity to depart for reasons unrelated to the heat.

Nevertheless, correctional officers might be more likely to leave their jobs in the summer months for reasons that are unrelated to the heat. Overall departures across the U.S. workforce follow a seasonal pattern of increased separations in the summer months, followed

4. See Section B in the Supplementary Information for further detail on how we construct this variable.

5. We do not know whether someone voluntarily or involuntarily leaves the workforce, only that they depart our dataset.

by decreased exits in the winter ("Making sense of seasonal adjustments to job quits — FRED Blog" 2023). As a result, our key test is not only whether departures increase in the summer months, but whether the magnitude of the difference is larger in prisons without air conditioning. In other words, we test whether the excessive heat caused by working in an uncooled prison in the summer leads to a greater increase in the probability of departing than working in a cooled prison in the summer.

$$depart_{ipm} = \alpha + \sum_{k=1}^{4} \left(\beta_k temp_{pm}^k \times no\ ac_p\right) + \sum_{k=1}^{4} \gamma_k temp_{pm}^k + \Omega \mathbf{X}_{pm} + \Phi \mathbf{Z}_{im} + \sigma_y + \rho_p + \epsilon_{ipm} \quad (1)$$

Formula 1 specifies our approach to testing for a moderating effect of air conditioning in the hottest months of the year. The dependent variable *depart* is a binary indicator for whether employee i working in prison p departs the TDCJ workforce in month m.[6] The main independent variable is the interaction between the monthly outside temperature in the county containing a correctional officer's prison of employment (*temp*) and whether the prison lacks full air conditioning (*no ac*). We measure temperature as the average monthly temperature (in Fahrenheit) of the county containing a prison.[7] Following other researchers (Mukherjee and Sanders 2021; Deschênes and Greenstone 2011; Barreca et al. 2016; Heutel, Miller, and Molitor 2021), we bin average temperature readings to pick up on nonlinear trends.[8] In particular, we create five bins by degrees: less than 60°F, 60-69°F, 70-79°F, and 80°F or higher.[9] While we include all values in the regression, we are particularly interested in the mean difference in the probability of departing when average temperatures are 80°F or higher, relative to less than 60°F.

6. Since we count instances where an employee leaves the TDCJ and then returns in a future month as a departure, the unit of analysis is an employment spell rather than an employee.

7. Temperature data downloaded from the National Centers for Environmental Information's "nClimDiv" dataset. Figure C.1 shows that counties with and without air conditioning have similar seasonal temperature fluctuations.

8. Binned temperature measures introduce the possibility of bias if outcomes correlate with locations' baseline temperatures (Jones et al. 2026). We are largely unconcerned about this issue in our design because of the similar baseline temperatures across prisons and lack of significant changes in baseline temperatures over our study period (See Figure C.1).

9. Figure C.2 in the Supplementary Information shows the distribution of our binned temperature variable. Figure C.3 shows the seasonal distribution of temperature readings over the course of the year.

We also include a series of individual-level, prison-level, and time controls in the model. The matrix $\mathbf{Z}$ includes controls for employees' race, age, squared age, logged monthly salary, years of experience,[10] and sex. We include prison-level fixed effects ρ to pick up on differences in prisons' overall size, location, security status, and age. (Since prisons' air conditioning status is also time-invariant, we exclude a main effect for *no ac.*) While our model controls for time-invariant differences across prisons, it is still possible that departures in prisons with or without air conditioning vary over the course of the year in ways correlated with outside temperatures. Therefore, we also include time-varying prison-level controls ($\mathbf{X}$), including the monthly unemployment rate of the county containing the prison, the percent change in the number of correctional employees working at the prison over the prior six months, the number of security staff working at the facility in the given month, and the logged number of incarcerated people in the facility in the given month. Officials in the state have pointed to rising oil and natural gas production during the beginning of our study period as contributing to understaffing in the TDCJ, and we therefore also control for the logged monthly amount of oil and natural gas produced in the county containing the given prison (Foxhall 2012; McCullough 2019a).[11] Finally, we include calendar year fixed effects (σ). We estimate the model using logistic regression and cluster all standard errors by prison.

The key identifying assumption in this design is that hot temperatures only have a differential impact on departures in prisons without air conditioning because of the excessive heat caused by the lack of air conditioning, not other unobservable factors. Potential threats arise if prisons lacking air conditioning differ in other ways that affect turnover in the hottest months of the year. For instance, if prisons without air conditioning lack modern ventilation systems, this could drive increased turnover in the summer months independent of air-conditioning status. While prison fixed effects control for time-invariant differences across facilities, over-time changes in prison maintenance across prisons with and without air conditioning could still arise over the course of our study period. Other potential identification threats are on the selection side. While we control for a number of observable individual-level variables, it might be the case that employees who opt into working in prisons without air

10. We control for employees' years of experience with fixed effects to pick up on nonlinearities.
11. Figure A.1 shows the distribution of oil and natural gas production over time.

conditioning are, at baseline, more likely to depart. While this could underscore the influence of air conditioning, it would be problematic if selection was due to another, unobservable feature of these prisons.

In addition to the numerous controls in our model, we address these possible threats to inference by specifying our model on various subsets of the TDCJ personnel data, running placebo tests, and conducting robustness checks. We run the main model using all employees, all frontline correctional officers, all frontline correctional officers with five or fewer years of experience with the TDCJ, and all frontline correctional officers with six or more years of experience. We expect our results to largely hold for newer, frontline employees. More experienced correctional employees will be much less likely to be affected by the heat. In addition, while we are predominantly interested in the differential impact of the hottest temperatures, relative to the coldest, we also run placebo tests that assess whether differences in turnover still exist across prisons with and without air conditioning in the spring or fall relative to the winter. Not rejecting the null of no difference would lend further support to air conditioning, not some other factor, driving any observed differences in the summer months. Finally, we also conduct robustness checks using alternative model specifications, which we discuss in greater detail in the Results section.

The models in the text use data from 2010 up to the onset of the COVID-19 pandemic. Estimates for the entire study period and the years after March 2020 are located in the Supplementary Information. While our results largely hold over the entire study period, the findings for the period following March 2020 strongly diverge from the findings in the pre-pandemic period. These differences could be the result of changing turnover dynamics related to heat and air conditioning, but we believe they more likely result from the considerable—and difficult to control for—effects of the COVID-19 pandemic on prisons. The COVID-19 pandemic triggered a series of dramatic and overlapping changes in prison administration, affecting nearly every dimension of day-to-day operations. Interactions with incarcerated individuals were altered by distancing protocols, movement restrictions, and heightened anxieties regarding sanitation and infection-control. These challenges, combined with heightened absenteeism, increased mandatory over time, and changing policies, resulted

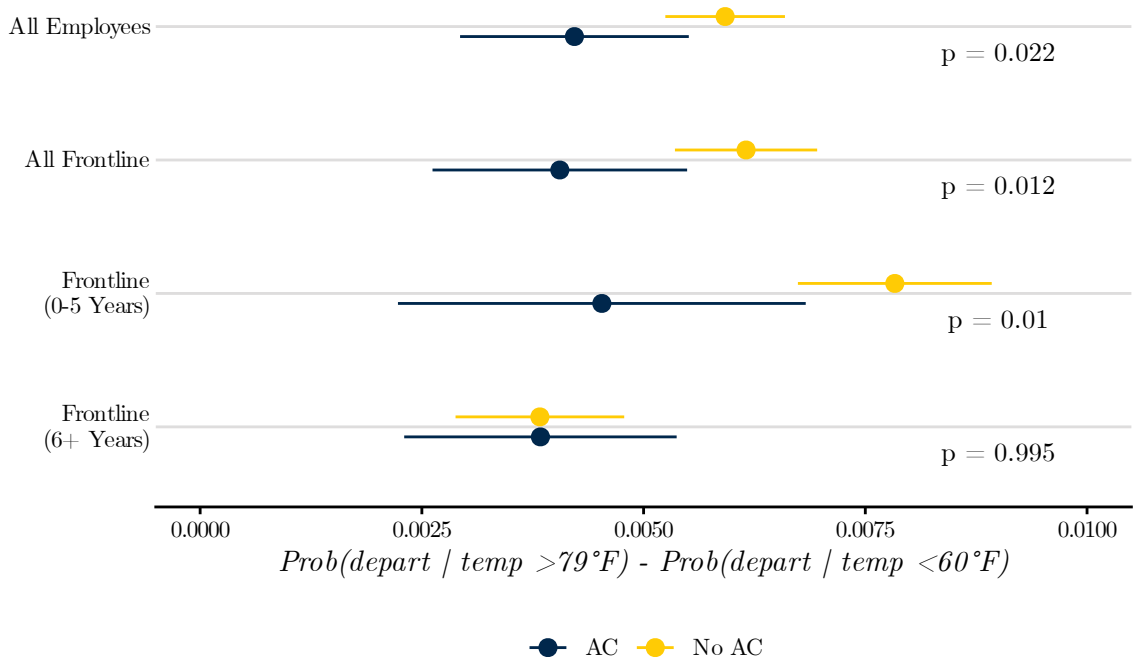
in considerable changes to correctional officers' jobs (Ferdik, Frogge, and Doggett 2022; Novisky et al. 2023; Schultz and Ricciardelli 2022). Consequently, we focus our attention on the pre-COVID period.

5 Results

Figure 4 displays our main results. Each point estimate shows the difference in the predicted probability of departing in the hottest months of the year (average temperature >79°F) relative to the coldest (average temperature <60°F) for employees working in, respectively, prisons with and without air conditioning. We compute the predicted probabilities using an observed-value approach (Hanmer and Ozan Kalkan 2013; Arel-Bundock, Greifer, and Heiss 2024). In other words, we stratify by prisons' air conditioning status and estimate the predicted probability of departing for every employee with the binned *temp* variable set to each of its respective values and all other controls set to their observed values.[12] Then, for employees in prisons with and without air conditioning, we estimate the difference in the mean predicted probability of departing when *temp* is set to its hottest and coldest values. Finally, we conduct a hypothesis test of no difference in the predicted probability of departing for employees working in, respectively, prisons with and without air conditioning. The p values from those hypothesis tests are displayed on the right side of Figure 4.

12. Due to computational constraints, we estimate the predicted probabilities using a 50% random sample of our dataset.

Figure 4 – Departing in the Summer More Likely in Prisons without Air Conditioning. Shows the difference in the predicted probability of departing when average monthly temperatures are >79°F versus <60°F within, respectively, prisons with and without air conditioning. The predicted probabilities are estimated using Formula 1 and the `marginaleffects R` package. The personnel data is subset to before the onset of the COVID-19 pandemic. Figure D.4 shows that our results largely hold for the full period from 2010-2023, and Figure D.5 shows results for after the onset of the pandemic. Figures D.2 and D.1 show that the findings largely hold using, respectively, more and less expansive definitions of prisons' air conditioning status.



Regardless of experience or air conditioning, correctional officers are more likely to depart in the hottest months of the year. However, besides the most experienced frontline correctional officers, employees working in prisons without air conditioning are considerably more likely to depart than their peers in air-conditioned prisons. Among all correctional employees (top row in Figure 4), those in facilities with and without air conditioning were, respectively, 0.42 and 0.59 percentage points more likely to depart, on average, during the hottest months of the year. The difference between these two estimates is statistically and substantively significant. Staff working in facilities without air conditioning were 40% more likely to de-

part in the summer months, compared to their peers in air-conditioned prisons ($p = 0.022$). As expected, the moderating influence of air conditioning on departures is strongest among newer, frontline correctional officers. Frontline correctional officers and frontline correctional officers with five or fewer years of experience were, respectively, 52% and 73% more likely to depart in the summer months if they worked in a prison without air conditioning. In contrast, as shown in the bottom row of Figure 4, there is no difference in the probability of departing in the summer months for employees with six or more years of experience working in prisons with and without air conditioning.

Figure 4 shows that the extreme heat in prisons without air conditioning during the summers leads to increased correctional officer turnover. If the extreme heat introduced by the lack of air conditioning is the actual source of this difference in turnover rates, then the difference should subside as outside temperatures decrease. Figure D.6 in the Supplementary Information shows that this is the case. The results suggest that similar differences in turnover exist across prisons with and without air conditioning when average temperatures are between 70-79°F, relative to <60°F. However, there is no significant difference in the probability of departing when temperatures are 60-69°F, relative to <60°F. This suggests that air conditioning continues to moderate employees' probability of departing when temperatures are warm in months like May, September, and October. However, as temperatures decline to cooler levels where air conditioning is not necessary, the differences in departure rates across prisons with and without air conditioning become insignificant. This underscores that our results are most likely being driven by changes in indoor temperatures caused by the lack of air conditioning rather than another unobserved difference across prisons.

The results in Figure 4 exploit variation in binned mean monthly temperature. These binned measures correlate strongly with the month of the year, meaning that our results are largely picking up on seasonal trends rather than marginal increases in monthly average temperature. This approach fits with other scholars' modeling approaches and picks up on the large seasonal temperature swings over the course of a year in Texas. Yet, as a robustness exercise, we re-estimate Formula 1 with the numeric, unbinned measure of average monthly temperature and monthly fixed effects in place of annual fixed effects. This is a much more

conservative test. The monthly fixed effects absorb seasonal fluctuations in temperature, leaving only deviations from mean temperatures as a source of temperature variation. In Figure D.3 in the Supplementary Information, we show that despite the conservative nature of the test, our results still hold using prisons with either full or no air conditioning. Results using more expansive definitions of prisons' air conditioning exposure are mixed but still point in the expected directions.

6 The Cost of Installing Air Conditioning

The results above suggest that installing air conditioning in TDCJ prisons would improve the retention of correctional officers. Among other potential benefits of installing air conditioning (e.g., improved safety of incarcerated people and reduced court costs from heat-related litigation), the TDCJ and state stand to gain from the reductions in employee turnover facilitated by installing air conditioning. Current high levels of employee turnover in the TDCJ interfere with productivity and safety, require increased spending on recruiting and training new hires,[13] and necessitate expensive mandatory overtime.[14] The TDCJ is aware of these costs and the potential benefits of installing air conditioning. The former Executive Director of the TDCJ testified in court that he "absolutely" wants the entire TDCJ system to be air conditioned (Pitman 2025 at p. 86).

Yet, installing air conditioning is expensive. As of August 2024, The TDCJ has approximately 95,000 uncooled beds in facilities with partial or no air conditioning. Many of these beds are located in prisons built well before the invention of modern heat mitigation technologies, complicating the construction process. In this section, we conduct back-of-envelope cost estimates of installing air conditioning throughout the TDCJ system. In order to provide context for these cost estimates, we also estimate the cost of an alternative, frequently used policy option for improving employee retention: raising salaries. We demonstrate that, relative to a salary increase that reduces turnover by the same amount as installing air conditioning, installing air conditioning is a relatively cost-effective policy for addressing

13. The state estimates that advertising for a correctional officer position and training a new hire costs approximately $10,000 (*Sunset Report* 2025).

14. The TDCJ spent $277 million on overtime in fiscal year 2023 (*Sunset Report* 2025).

understaffing.

There is no consensus about how much it would cost to install air conditioning throughout all TDCJ facilities. As a result, we use estimates from the TDCJ, legislative proposals, and prior air-conditioning projects to generate four different cost scenarios. The first scenario is from a 2023 TDCJ estimate of the cost of implementing a legislative bill mandating air-conditioning in all TDCJ facilities. The agency estimated that installation would cost over $1.1 billion, with annual maintenance costs totaling $20 million (McGinty 2023). This estimate is largely in line with an earlier estimate from 2019 that also placed the cost of installation at over $1 billion (McCullough 2019b). Critics of the 2019 estimates suggested that the agency had a history of over-estimating the costs of air conditioning. For instance, during a lawsuit over excessive heat in the Pack Unit the TDCJ initially stated that adding air conditioning to the facility would cost $20 million, but costs ended up totaling $4 million.

The second and third scenarios use appropriations and actual cost data to estimate installation costs per bed. The TDCJ was recently appropriated $118 million to install air conditioning covering 18,000 beds, at a cost of $6,556 per bed. Applying the $6,556/bed cost to the 95,000 beds that remain uncooled, installing air conditioning in this scenario would cost approximately $623 million. The third scenario uses actual cost data from three construction projects. In 2018, the TDCJ added air conditioning to the Pack and Hodge units due to a court order, at a cost of, respectively, $4 million and $2 million (McCullough 2018b). Likewise, in fiscal year 2021, the TDCJ air-conditioned 3,407 beds at a cost of $13.4 million. Applying the cost per bed of these three projects (in 2025 dollars) generates a cost estimate of $404 million. Finally, the fourth scenario uses amounts from a 2025 Texas House bill that called for the state to spend $300 million to install air conditioning in all TDCJ facilities over the next seven years (H.B. 3006 2025).

Table 1 overviews the cost of installing air conditioning in the four scenarios. Outside of the first funding scenario where the TDCJ provided an estimate of the cost of annual maintenance, we assign the annual operating cost C_{oper} to be 1.6% of the initial capital cost of installation, which is the ratio of annual maintenance costs to upfront air-conditioning

Table 1 – Cost Scenarios. Shows the upfront installation and annual maintenance costs of installing air conditioning under four scenarios. All dollar amounts are in millions of 2025 dollars. Under scenarios 2-4, annual maintenance costs are estimated as 1.6% of the initial installation costs. This is the ratio of annual to initial costs in a recent report outlining the costs of installing air conditioning in prisons operated by the Florida Department of Corrections.

Scenario	Description	Initial Cost (C_{cap})	Annual Cost (C_{oper})
1	2023 TDCJ Estimate	\$1,125	\$20
2	2025 Appropriation	\$623	\$10
3	Installations	\$404	\$6
4	H.B 3006	\$300	\$5

installation costs in a recent report prepared for the Florida Department of Corrections (KPMG 2023). We annualize these initial and annual costs using Formula 3. Solving the equation requires selecting a discount rate and expected lifespan of the air conditioning system. Rather than selecting a single value for each of these parameters, we estimate C_{annual} with inputs $\mathcal{R} \times \mathcal{L}$, where:

$$\begin{aligned} \mathcal{L} &= \{15, 20, 25, 30, 35, 40\}, \quad L \in \mathcal{L} \\ \mathcal{R} &= \{0.03, 0.04, 0.05, 0.06, 0.07\}, \quad r \in \mathcal{R} \end{aligned} \tag{2}$$

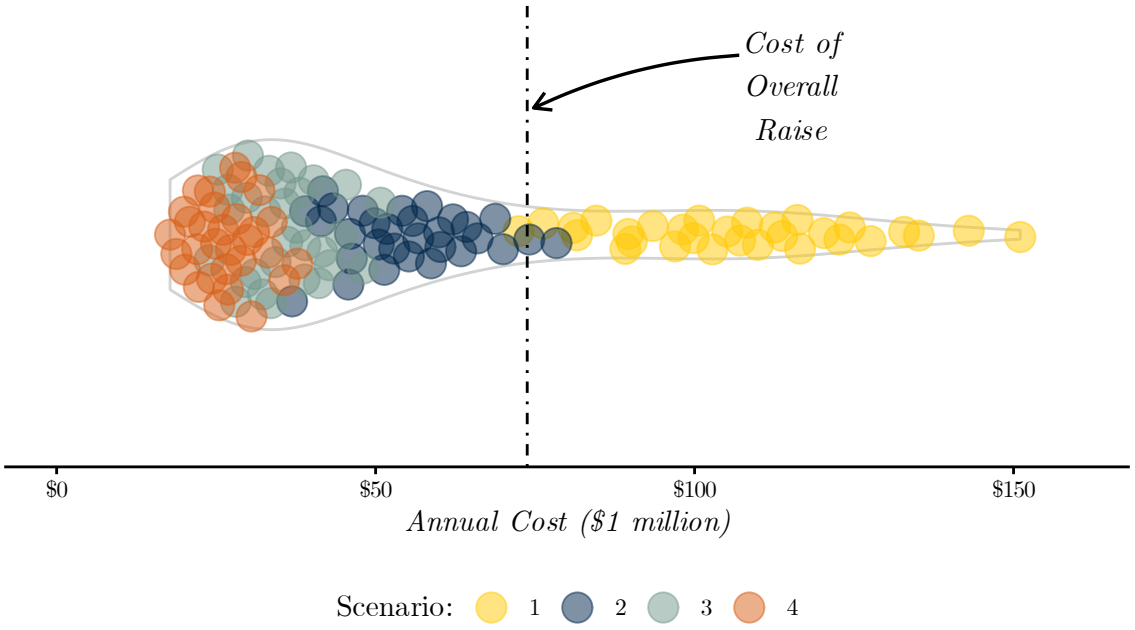
$$C_{\text{annual}} = \underbrace{C_{\text{cap}} \left(\frac{r(1+r)^L}{(1+r)^L - 1} \right)}_{\text{Annualized capital cost}} + \underbrace{C_{\text{oper}}}_{\text{Annual operating cost}} \tag{3}$$

$$\text{where } r = \text{discount rate}, \quad L = \text{system lifespan (years)}$$

In order to benchmark our cost estimates, we estimate the cost of raising employee salaries by the amount necessary to reduce turnover by the same amount as installing air conditioning. Salary raises are a favored policy tool to improve employee retention in the TDCJ. The agency has described the 15% pay raise given to correctional officers in August 2022 (shown in Figure 3) as having had "the most significant impact on Correctional Officer staffing levels compared to any other hiring initiative" (Collier 2024a). Since 2022, correctional staff in the TDCJ have received four raises, increasing salaries by 40% over the period.

We estimate the cost of raising employee salaries by the amount necessary to generate the same reduction in turnover among frontline correctional officers as installing air conditioning. Since our model in Figure 4 is an estimate of the seasonal treatment effect of air conditioning, we estimate the annual difference in turnover across prisons with and without air conditioning. Using values for all frontline correctional officers from Figures 4 and D.6, we calculate the mean difference in seasonal turnover across prisons with and without air conditioning weighted by the share of months in our dataset that fall within each temperature bin.[15] Using this estimate and the average marginal effect of increasing monthly salaries on the probability of departing (estimated via Formula 1), we find that raising salaries by 7.6% correlates with the same reduction in turnover as installing air conditioning.[16]

Figure 5 – Annualized Cost of Air Conditioning Similar to Equivalent Salary Raise. Shows the distribution of the estimated cost of installing air conditioning in all TDCJ facilities, by upfront cost scenario and different assumptions about system lifespan and the discount rate. The dashed line denotes the estimated cost of raising salaries of frontline correctional officers in prisons with partial or no air conditioning by 7.6%. The dotted line displays the estimated cost of only raising salaries for the least-experienced employees. All amounts are in 2025 dollars.



15. We assume zero difference in turnover between prisons with and without air conditioning during the coldest months of the year.

16. See Section E in the Supplementary Information for more details on these calculations.

The dashed line in Figure 5 shows the cost of raising salaries of all frontline correctional officers in prisons without full air conditioning by 7.6%. The colored dots display the distribution of the annualized cost of installing air conditioning. The multiple dots within each scenario reflect the different assumptions regarding discount rates and the lifespan of the air conditioning system. Overall, the annual cost of installing air conditioning varies widely both within and across scenarios, ranging from around $25 million to $150 million annually (median = $43 million, mean = $56 million). In comparison, raising salaries for all frontline correctional officers in prisons lacking full air conditioning costs approximately $81 million per year. Besides the cost estimates generated from Scenario 4, raising employee salaries is more expensive than nearly all estimates of the cost of installing air conditioning.

Of course, caveats exist with these back-of-envelope cost estimates. For one, the benefits of air conditioning are not immediate. Unlike salary raises, which can be quickly implemented, approving the installation of air conditioning is not the same as actually cooling prisons. The construction process means that the benefits of air conditioning on turnover take time to accrue, which we do not account for in our estimates. In addition, we also treat the effect of air conditioning as constant across prisons that lack full air conditioning. This might not be the case, as the marginal benefits on turnover of, for example, installing 100 cooled beds might be less in a prison that is already nearly fully air-conditioned. While we do not account for this here, Figure 1 shows that most prisons without full air conditioning have minimal coverage, likely dulling the implications of this concern for our cost estimates.

7 Discussion

Reformers have long framed the lack of air conditioning in many U.S. prisons as a moral and constitutional issue. We offer a different, albeit complementary, perspective, arguing that the poor working conditions are also a workforce issue. The street-level bureaucrats who work in prisons, like the incarcerated people they supervise, are negatively affected by excessive heat. However, unlike incarcerated people, correctional officers have the option to leave, further exacerbating the dangerously high levels of turnover and understaffing that plague the American carceral state.

Using personnel and air conditioning data from the largest state prison system in the U.S., we show that excessive heat caused by a lack of air conditioning contributes to correctional officer turnover. Before the onset of COVID-19, correctional officers in prisons without air conditioning were 40% more likely to depart the TDCJ workforce in the summer months than their peers in cooled facilities experiencing similar outside temperatures. While we add to a growing literature on how excessive heat fosters dangerous conditions within U.S. prisons, we innovate by both focusing on correctional officers and examining the cost-effectiveness of investing in air conditioning. We demonstrate that, when it comes to reducing correctional officer turnover, installing air conditioning imposes similar or even less costs than raising employee salaries. The initial investments of adding air conditioning are high, but given the size of the correctional officer workforce in Texas and many other states, raising employee salaries also comes at a considerable cost. Furthermore, adding air conditioning has other financial benefits not taken into account in our model. For instance, with air conditioning installed, states no longer need to pay to defend themselves in court from heat-related lawsuits brought by incarcerated people.

Our work has a few key limitations, which should be improved upon in future work. First, additional work should analyze the effects of heat on turnover during and after the COVID-19 pandemic. We find that our results are largely isolated to before the onset of the COVID-19 pandemic. There are a few possible reasons for this that deserve additional study in future work. The pandemic brought unprecedented challenges to prisons, many of which were particularly affected by the virus. As a result, new unobserved confounders that are difficult to control for may be biasing our results and masking an underlying relationship between air conditioning and departing. The lack of considerable data from "after" the pandemic makes it difficult, in this study, to fully disentangle the possible reasons for the variation in our findings over time.

Future work should also investigate other possible responses to excessive heat among correctional officers and other types of street-level bureaucrats. Exiting the workforce is one, particularly extreme, measure that correctional workers can take in the wake of poor conditions. They can also take vacation days or be forced to work additional overtime shifts

to cover vacancies. Future research should incorporate these other dependent variables. In addition, while Texas is a particularly salient case for studying the relationship between heat and correctional officer turnover, correctional officers in other states might be even more responsive to heat. Climate change is increasing the number of hot summer days across the country in places where people are less used to excessive heat (Tuholske et al. 2024). As a result, correctional officers outside of the South are beginning to face increasing and unprecedented heat-related difficulties.

Overall, our findings underscore the need for additional investigations into how street-level bureaucrats' physical working environments affect their willingness to remain in their jobs. While the case of excessive heat in Texas' prisons is a particularly extreme case of poor working conditions, influential frontline public employees across the country work in substandard settings due to underinvestments, excessive heat, and other factors. By demonstrating the relationship between poor conditions and public sector turnover, our study demonstrates the importance of taking seriously the on-the-job safety and comfort of frontline public employees.

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Supplementary Information

Excessive Heat and Correctional Officer Turnover: Evidence from Texas

A TDCJ Facilities **SI-1**

B Defining Employee Departures **SI-1**

C Heat in Texas **SI-2**

D Additional Results and Robustness Checks **SI-6**

E Cost Analysis **SI-12**

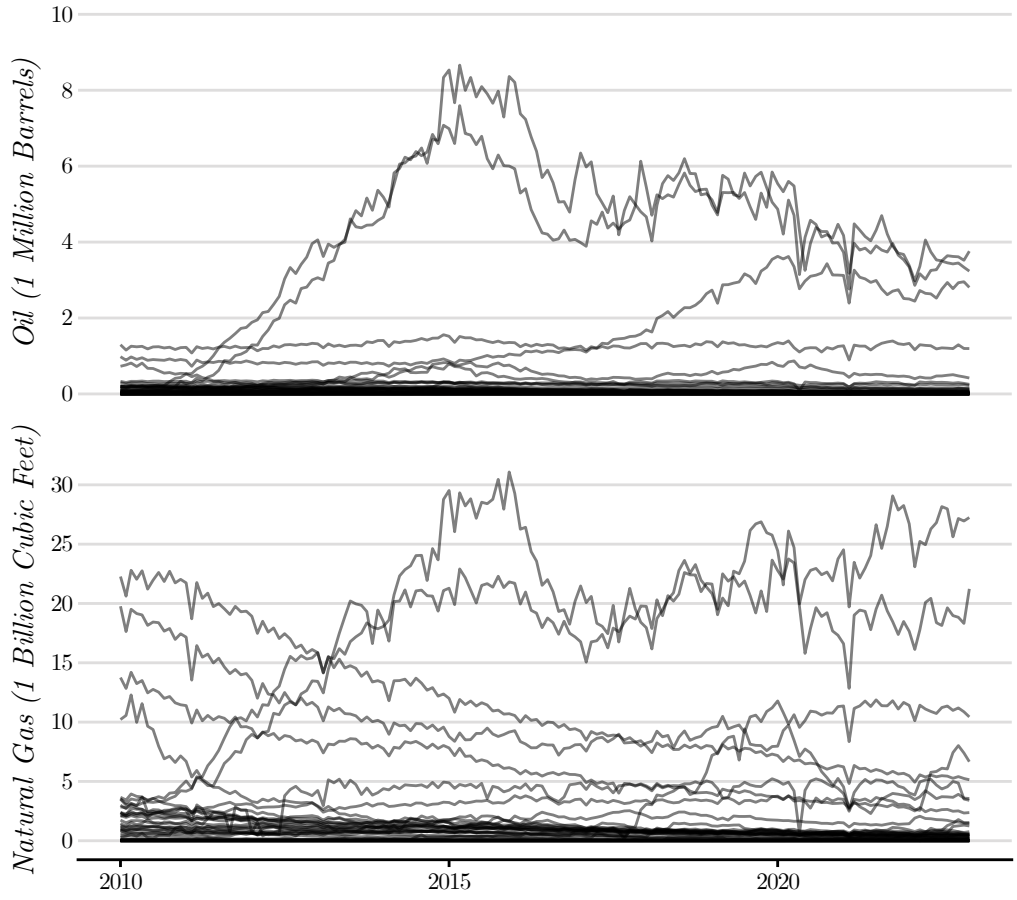
E.1 Estimating the Commensurate Percent Salary Raise SI-12

E.2 Estimating the Cost of the Percent Salary Raise SI-13

A TDCJ Facilities

Figure A.1 shows the monthly oil and natural gas production within each of the counties that includes a TDCJ prison included in our main results (i.e., Figure 4).

Figure A.1 – Monthly Oil and Natural Gas Production, by County. Shows monthly oil and natural gas production in counties with TDCJ facilities that have either $> 90\%$ or $< 10\%$ air conditioning coverage.



B Defining Employee Departures

The dependent variable in all analyses is whether an employee departs the TDCJ in a given month. We use information from two different sources to define an employee's month of departure. First, via the personnel dataset, it is straightforward to find the last month of observed continuous employment (i.e., the end of an employee's spell of employment with the TDCJ). However, since the personnel data is a snapshot of all TDCJ security employees on the last day of the given month, it is unclear whether someone actually departed on the last day of the month or at some point before the last day of the next month. The second source

of information is an additional dataset provided to us by the TDCJ that lists all employees who departed during our study period and their actual separation date (which we truncate to the actual month of separation). While more precise, these data do not include instances of employees moving from security to non-security (e.g., case manager) roles within the TDCJ. Despite their differences, the two data sources are largely similar. Employees' last observed month of employment and actual month of separation have a Jaccard index value of 0.89.

Given the benefits and drawbacks of both data sources, we use a two-pronged approach to define employees' month of departure. First, via the additional data provided to us by the TDCJ, we merged in the actual separation date for all employees who departed during our study period. Second, if an employee spell ends before the last month that we have data (December 2022) without a separation, we deem it to be an instance of an employee moving to a non-security role within the TDCJ and assign it as a departure.

C Heat in Texas

Figures C.1, C.2, and C.3 show the temperature distribution across seasons and TDCJ prisons. Figure C.1 shows that average temperatures vary significantly by month and are nearly identical in counties with prisons with and without air conditioning. Figure C.2 shows the distribution of our binned temperature variable. It shows that facility-month observations are more likely to have average temperatures lower than 60°F or greater than 79°F. It also shows little difference in outside temperature across prisons with and without air conditioning. Finally, Figure C.3 shows the monthly distribution of the binned temperature variable. Most of the highest-temperature months occur in June, July, and August. The coolest months largely occur in November, December, January, and February.

Figure C.1 – Mean Monthly Temperature of Counties with Prisons, by AC Status. Shows the mean monthly temperature of counties containing TDCJ facilities that have either $> 90\%$ or $< 10\%$ air conditioning coverage.

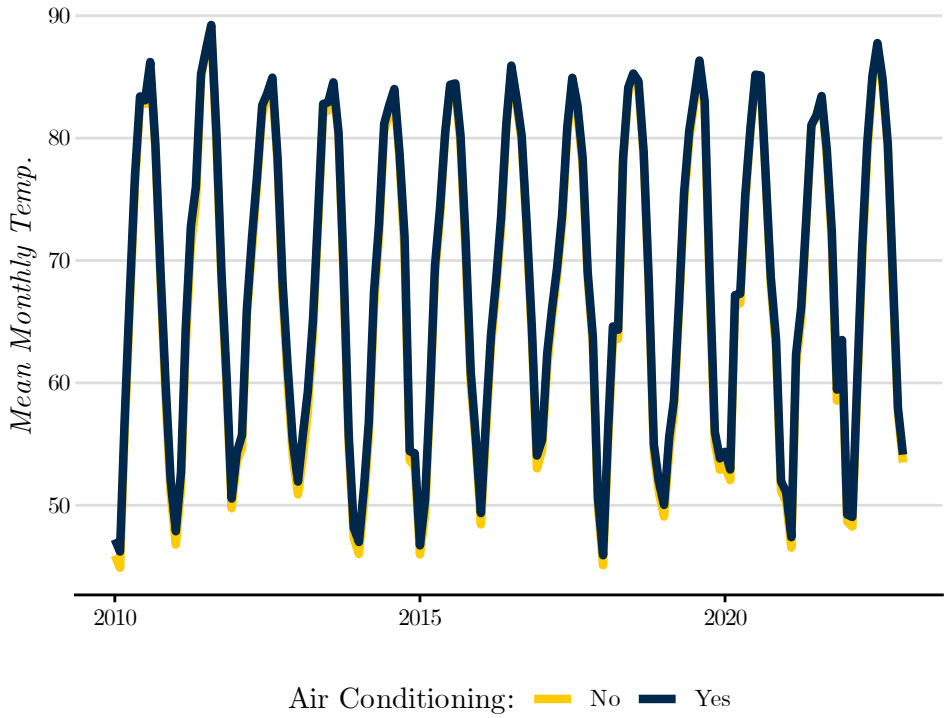


Figure C.2 – Distribution of Facility-Months by Temperature Bin. Shows the number of facility-months that fall in each of the temperature bins.

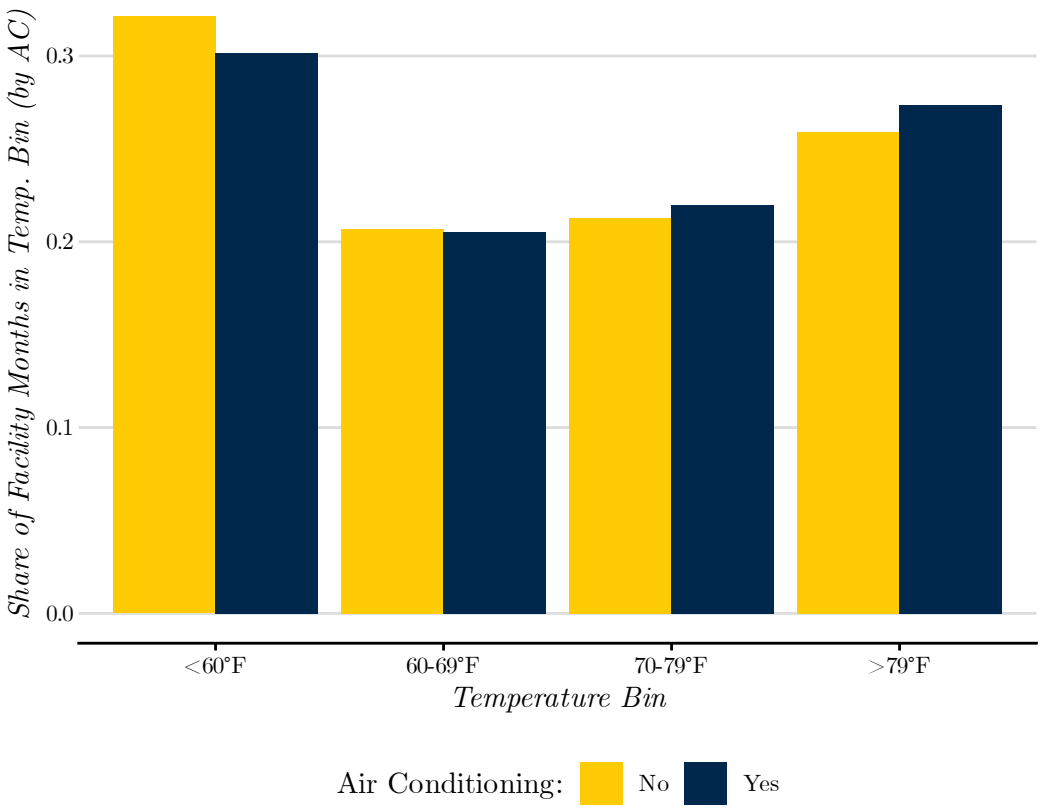
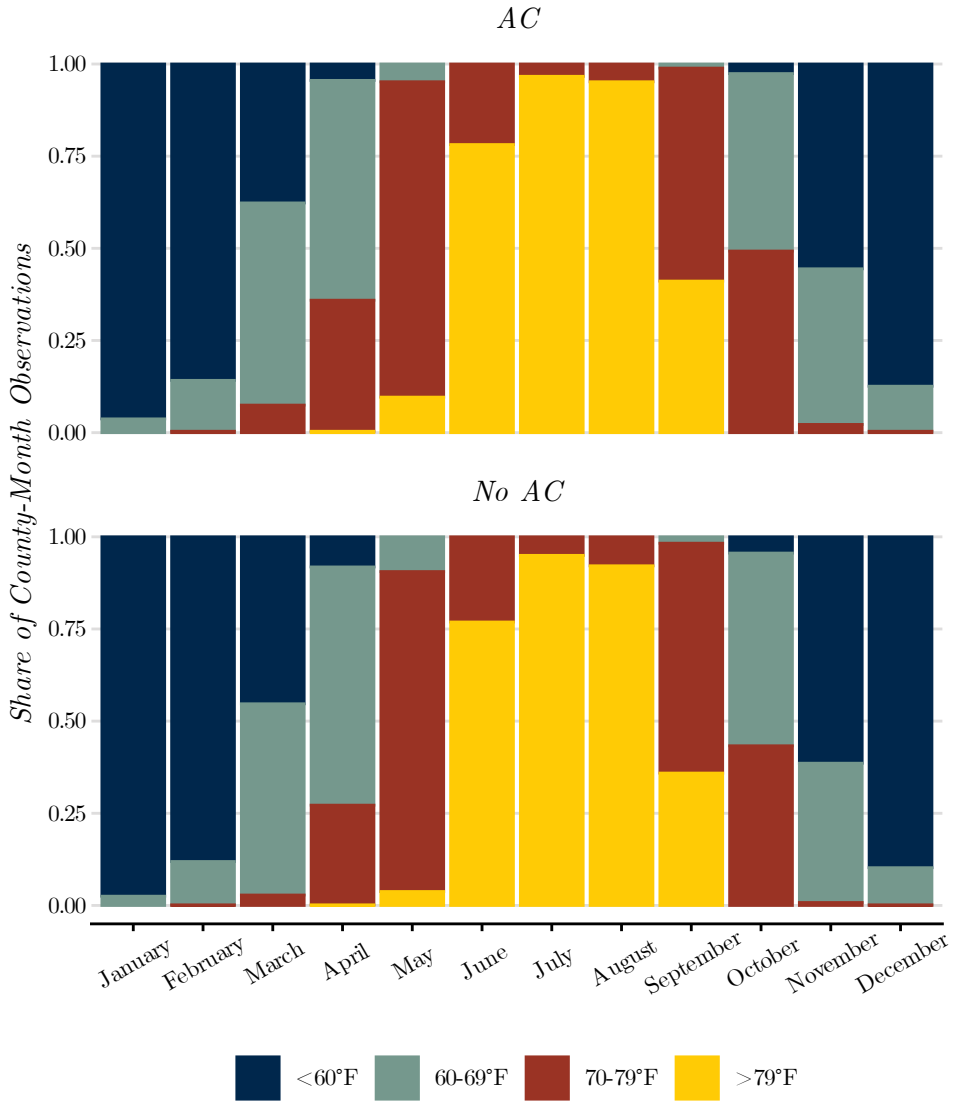


Figure C.3 – Distribution of Temperature by Month of Year. Shows the share of temperature readings by month of the year. The data includes all county-month observations from 2010-2023 for TDCJ facilities that have either $> 90\%$ or $< 10\%$ air conditioning coverage.



D Additional Results and Robustness Checks

This section shows that our main results in Figure 4 are largely robust to alternative definitions of which prisons do and do not have air conditioning (Figures D.1 and D.2), the inclusion of all data from 2010-2023 (Figure D.4), and an alternative model specification that includes the unbinned measure of average monthly temperature and monthly fixed effects (Figure D.3). In addition, Figure D.6 shows that our main finding of a difference in turnover across prisons with and without air conditioning dissipates as outside temperatures decrease. Finally, as discussed in the text, Figure D.5 shows that our main results do not hold for the period following the onset of the COVID-19 pandemic.

Figure D.1 – Replicates Figure 4 Using Only Prisons with Full or No Air-Conditioning. Shows the results of re-estimating the results in Figure 4 using an alternative definition of a prison having air-conditioning whereby a prison is classified as (not) being air-conditioned if the share of its beds with air conditioning is equal to 1 (0).

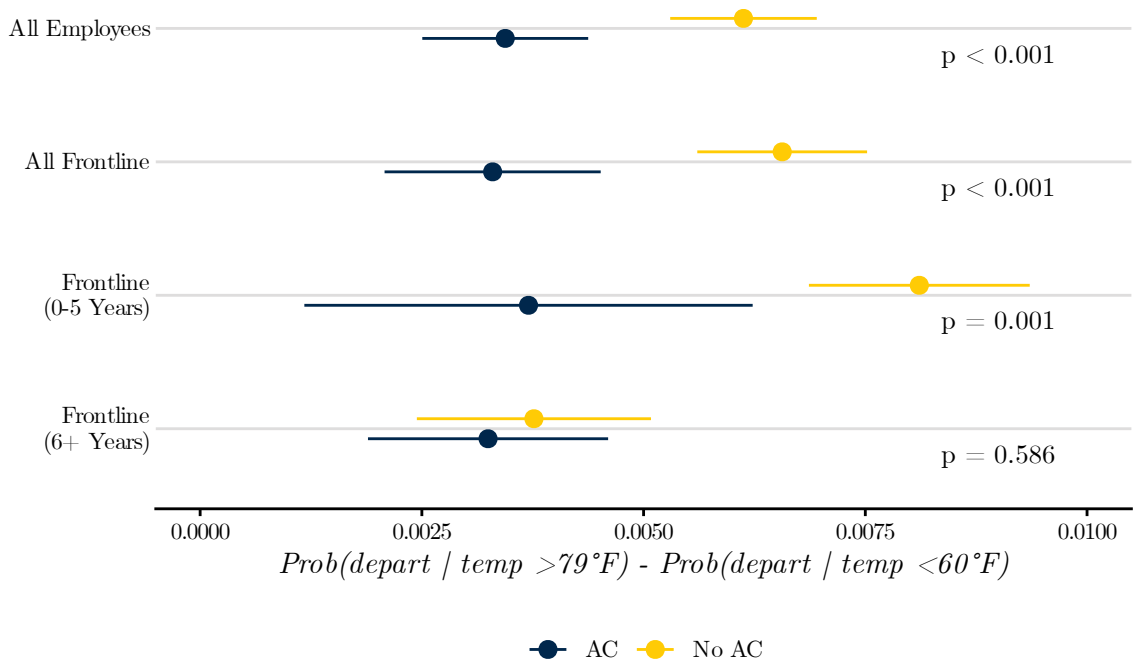


Figure D.2 – Replicates Figure 4 Using All Prisons. Shows the results of re-estimating the results in Figure 4 using an alternative definition of a prison having air-conditioning whereby a prison is classified as (not) being air-conditioned if the share of its beds with air conditioning is greater than (less than) 0.5.

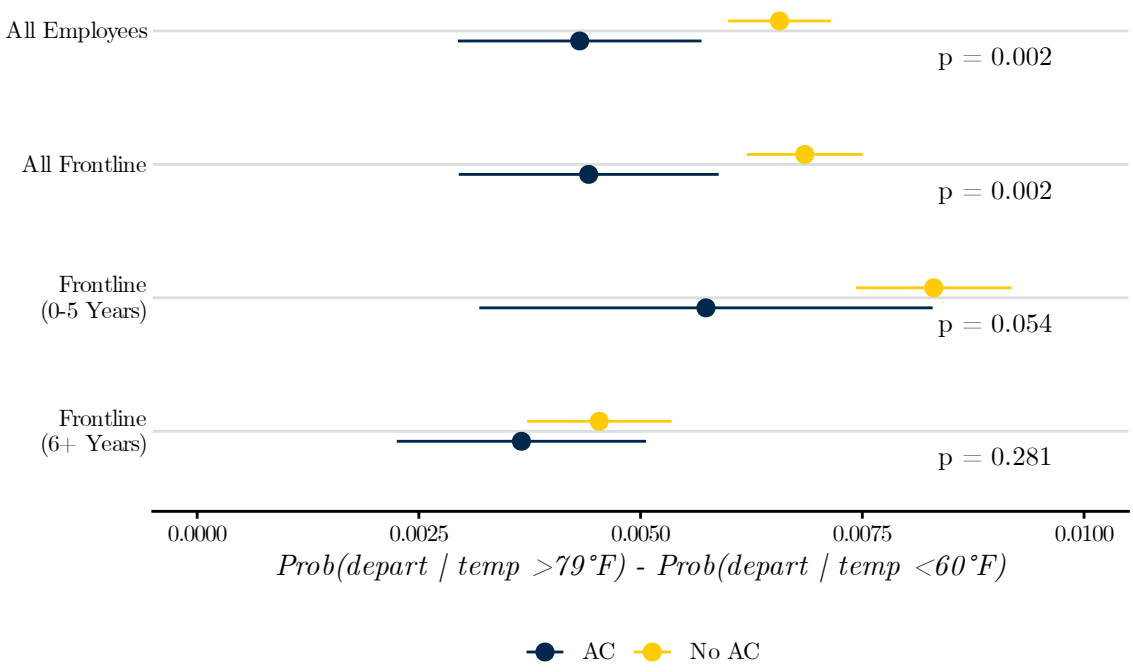


Figure D.3 – Mixed but Suggestive Moderating Effect of Air Conditioning Using Monthly Fixed Effects. Shows the results of an alternative version of Formula 1 that replaces the binned temperature variable with the unbinned, numeric average monthly temperature and replaces the annual fixed effects with monthly fixed effects. As such, each point shows the difference in the average marginal effect of increasing temperature in prisons without and with air conditioning. The top facet defines a prison as (not) having air conditioning if the share of beds with air conditioning exceeds (is less than) 0.5. The middle facet defines a prison as (not) having air conditioning if the share of beds with air conditioning exceeds (is less than) 0.9 (0.1). The bottom facet defines a prison as (not) having air conditioning if the share of beds with air conditioning is 1 (0).

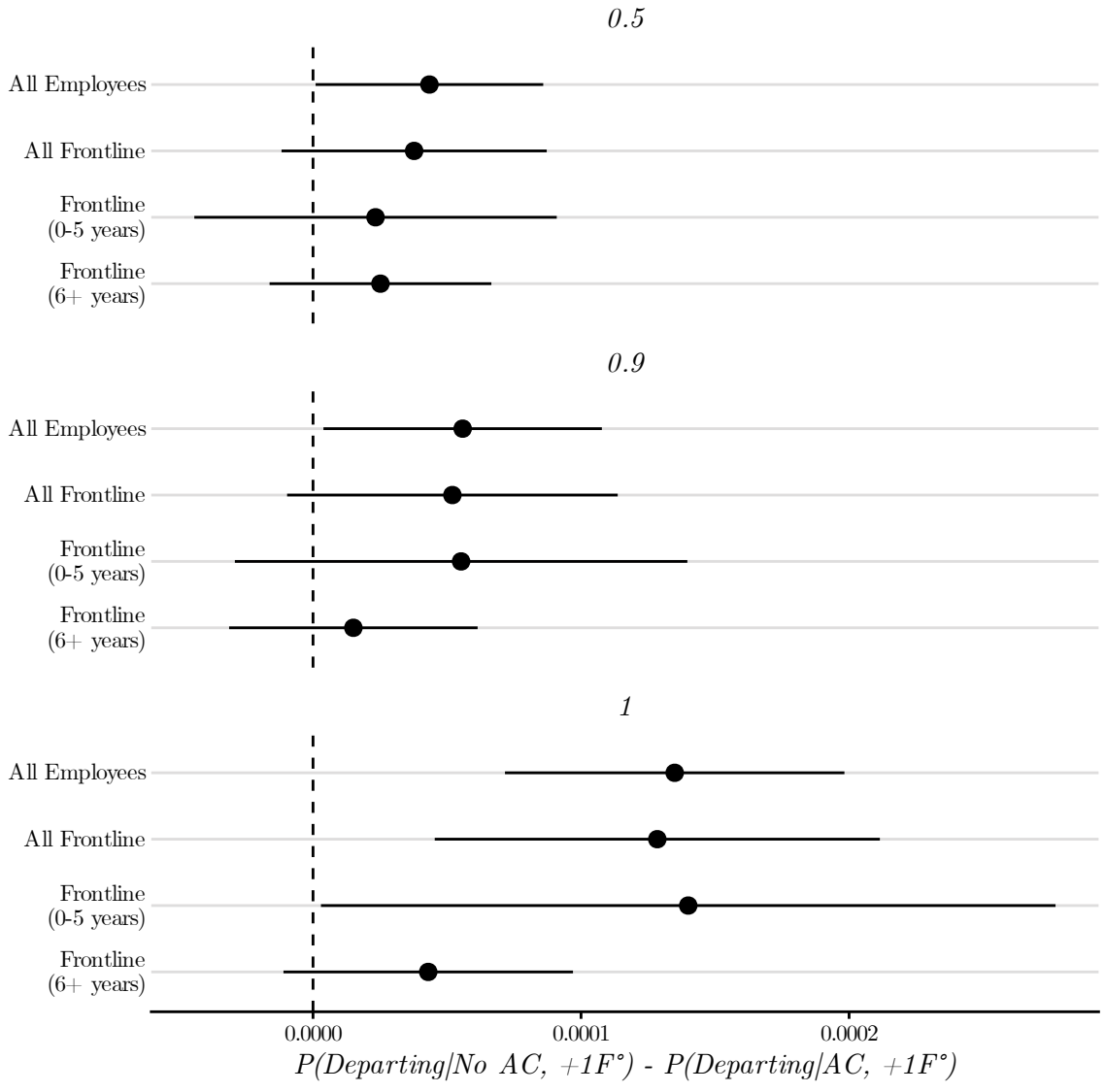


Figure D.4 – Departures in the Summer More Common in Prisons without Air Conditioning (Full Timespan). Replicates the estimates in Figure 4 using data from both before and after the onset of the COVID-19 pandemic.

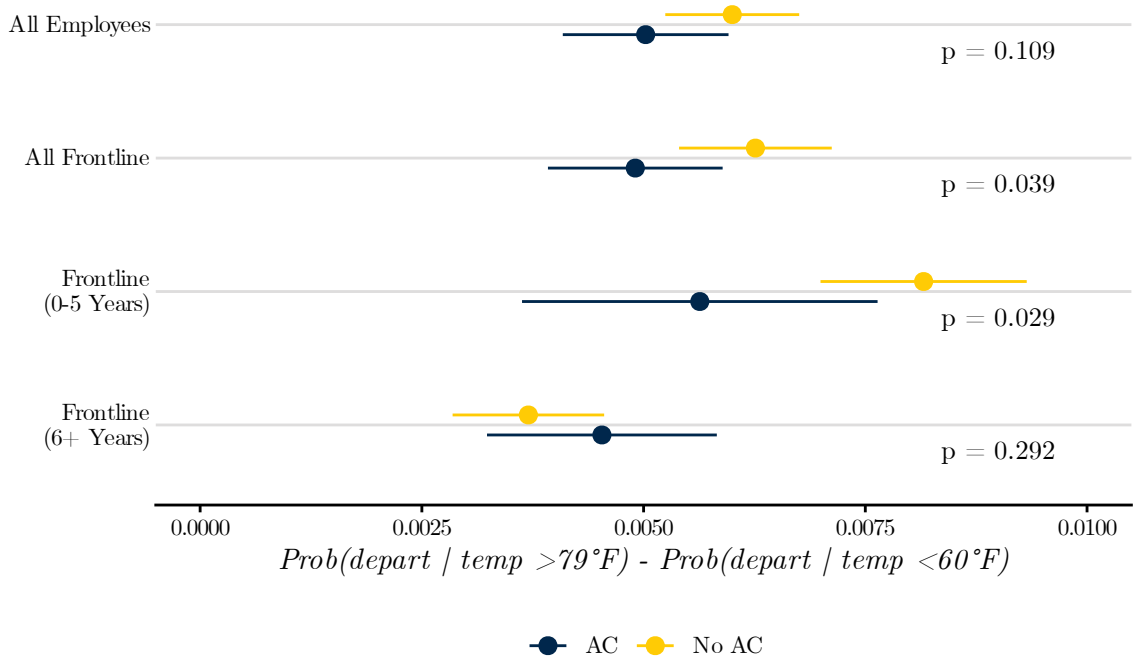


Figure D.5 – After the Onset of the COVID-19 Pandemic, Departures in the Summer Are Not More Common in Prisons without Air Conditioning. Replicates the estimates in Figure 4 using data from after the onset of the COVID-19 pandemic.

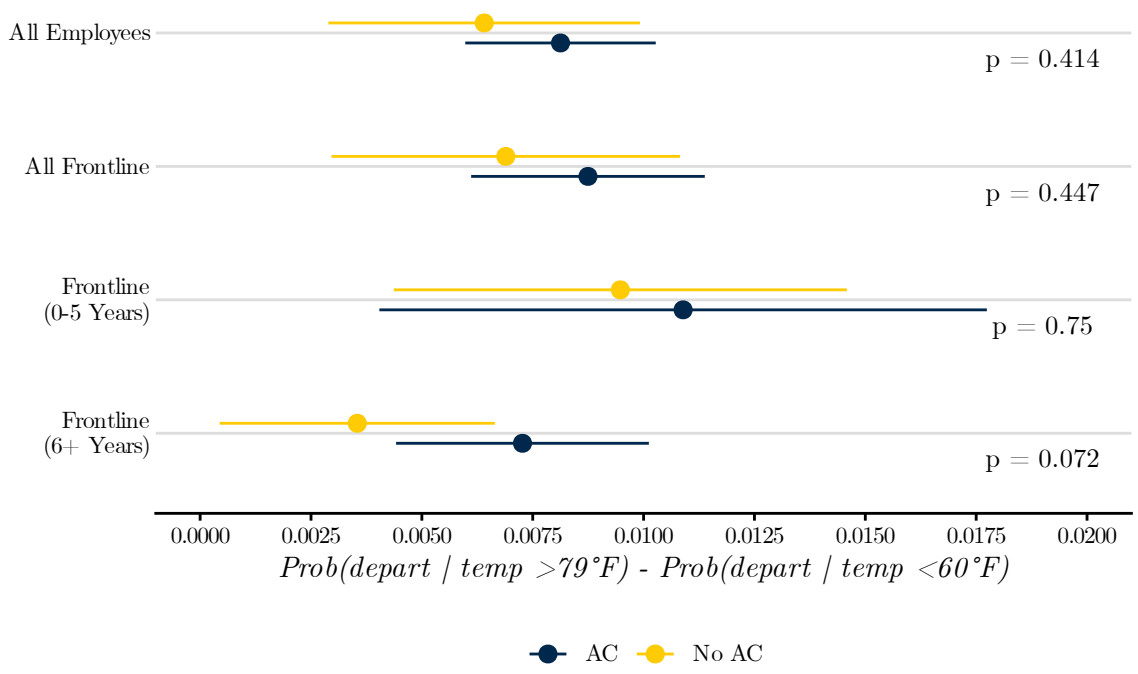
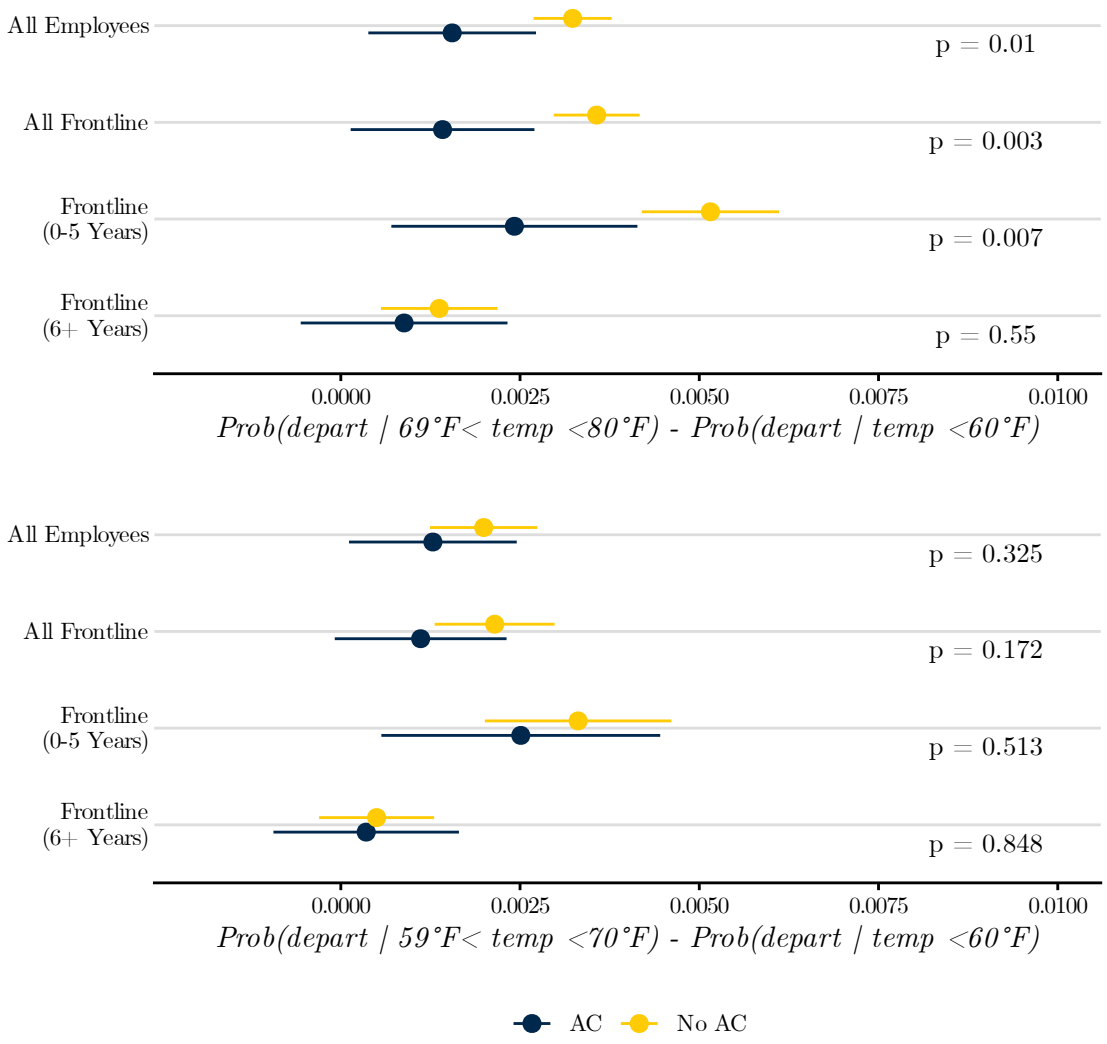


Figure D.6 – Employees in Prisons Without Air Conditioning More Likely to Depart When Average Temperatures are 70-79°but not 60-69°. Replicates the estimates in Figure 4 with using lower temperature levels. The top (bottom) panel shows the difference in the probability of departing when average temperatures are 70-79°(60-69°) relative to the baseline category of less than 60°.



E Cost Analysis

In this section we describe how we estimate the cost of raising employee salaries by the amount necessary to reduce turnover by the same amount as installing air conditioning. We estimate the cost of raising salaries for all frontline correctional officers employed in prisons with partial or no air conditioning by the percent necessary to reduce turnover by the same amount as installing air conditioning.

E.1 Estimating the Commensurate Percent Salary Raise

We first estimate the percent change in salary necessary to reduce the probability of departing in a given month by the same amount as installing air conditioning.

The annual predicted probability of a frontline correctional officer departing at baseline and with air conditioning installed is defined as a function of the baseline monthly probability of departing (p_m) and the difference in the probability of departing under air conditioning (Δ_{ac}):

$$p_a = 1 - (1 - p_m)^{12}$$
$$p_a^{ac} = 1 - (1 - (p_m + \Delta_{ac}))^{12}$$

We are interested in solving the following identity, where p_a^x is the annual predicted probability of departing when monthly salaries increase by x percent.

$$p_a^x - p_a = p_a^{ac} - p_a$$

Where:

$$p_a^x = 1 - (1 - (p_m + x \times \Delta_{salary}))^{12}$$

Rearranging terms:

$$0 = (1 - (1 - (p_m + \Delta_{ac}))^{12}) - (1 - (1 - (p_m + x \times \Delta_{salary}))^{12})$$

We solve for x using three inputs, which are shown below in Table E.1. The estimates of the change in the probability of departing when, respectively, monthly salaries increase by one unit and air conditioning is present are extracted from the models underlying Figure 4. p_m is the mean probability of departing in a given month for all frontline correctional officers before the onset of COVID-19.

Table E.1 – Inputs and Output of Percent Change in Salary. Shows the inputs used to define the percent change in monthly salary necessary to generate the same reduction the probability of departing as installing air conditioning. The percent change estimate is shown in the final column. Note that we multiply the values for Δ_{ac} by -1 because the models estimate the effect of no air conditioning. We divide the values for Δ_{salary} by 100 because they are on the log scale. Dividing by 100 gets the Δ_{salary} estimate to show the effect of increasing salary by 1%.

	Inputs			
	p_m	Δ_{salary}	Δ_{ac}	x
All frontline COs	0.021	-0.016/ 100	-1×0.0012	7.6

E.2 Estimating the Cost of the Percent Salary Raise

Now that we know the percent change in monthly salary necessary to reduce turnover by the same amount as installing air conditioning, we estimate the annual cost of the raises in inflation-adjusted 2025 dollars:

$$raise_a = cost_m \times scale \times 12 \times x \div 100$$

Where $cost_m$ is the total salary paid to frontline correctional officers in prisons with partial or no air conditioning in January 2023 (the last month we have data) and x is the percent raise necessary to match the effect of air conditioning on turnover. *scale* accounts for recent increases in salaries and staffing since January 2023. A recent report from the TDCJ notes that 18,757 frontline correctional officers work in the state (an increase of 9 percent from the total number in our data in January 2023). Likewise, according to a recent statement by the TDCJ, salaries have increased 40% since 2022, including a 15% raise in August 2022. Table E.2 lists the input values and corresponding estimates of the annual cost of the salary raises. As a spot check, we note that the TDCJ's budget request to the legislature for Fiscal Years 2026-2027 estimates that raising salaries for all correctional officers (including those employed in air-conditioned facilities) by 10% costs $165 million per year (Collier 2024b), which is largely in line, proportionally, with our estimate of the cost of a 7.6% raise for employees in facilities with no or partial air conditioning.

Table E.2 – Cost of Salary Raises. Shows the inputs used to estimate the cost of the salary raises necessary to generate the same reduction the probability of departing as installing air conditioning. All dollar amounts are in millions of 2025 dollars and are calculated only for frontline correctional officers working in prisons with partial or no air conditioning.

	Inputs			
	$cost_m$	*scale*	x	$raise_a$
All frontline COs	$59.2	1.09×1.25	7.6	$81